

Data Science in Medical Imaging

Coursework Minor A2

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1 Module 1: Tomographic reconstruction

1.1 Exercise 1.1: Reducing dose

(a) Setup

The CT slice was converted from RGB to grayscale and scaled by a factor of 100 before simulation. The final working image was 512×512 pixels with intensities roughly in the range $[0, 0.01]$.

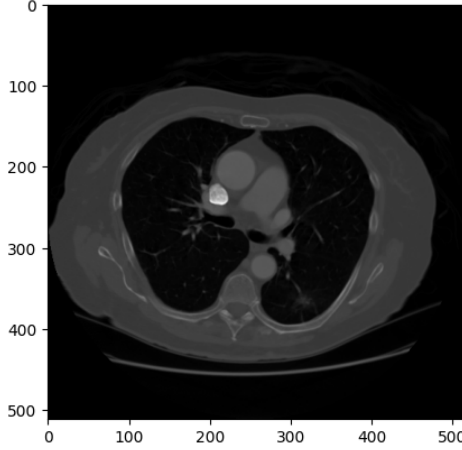


Figure 1: Input CT image used for Module 1.

(b) Noise simulation

Nine sinograms were generated for $I_0 \in \{10^2, 10^3, 10^5\}$ and $N_\theta \in \{20, 90, 360\}$ using Poisson plus Gaussian noise ($\mu = 0$, $\sigma = 0.05$). Figure 2 shows that raising I_0 mainly reduces graininess, whereas at 20 viewing angles the data is already clearly limited by undersampling.

(c) Reconstructions: FBP vs gradient descent (SIRT)

Figures 3 and 4, with Tables 1 and 2, show that FBP is more sensitive to sparse sampling and low dose. At $(20, 10^2)$ it shows strong streaking, RMSE 0.0125, SSIM 0.1813, and a negative PSNR of -1.97 dB, which means the reconstruction error exceeds the full dynamic range of the original image. Gradient descent on the same setting has RMSE 0.0007, PSNR 22.69 dB and SSIM 0.3316. Across eight of the nine settings GD has a lower RMSE and higher PSNR and SSIM than FBP, and the gap is largest in the weakest-data cases. However, it is also about $110\times$ slower (244.14 s versus 2.23 s) and the images are softer with some of the fine detail being smoothed out. This matters in the best case, $(360, 10^5)$, where FBP actually gives higher PSNR and SSIM and visibly sharper edges.

In a clinical setting, FBP stays useful when scan quality is high and fast reconstruction matters, while gradient descent is more helpful when dose is reduced because it can keep the image usable at lower exposure. Importantly though, at the lowest dose and fewest angles the blur may hide small but important details.

(d) Improving image quality

Two potential improvements are total variation regularisation, which can reduce streaks while keeping edges clearer, and a Poisson-based statistical reconstruction, as the photon counts follow a Poisson model. Also, a learned prior such as a plug-and-play denoiser may also help, although it would need much stronger validation before clinical use.

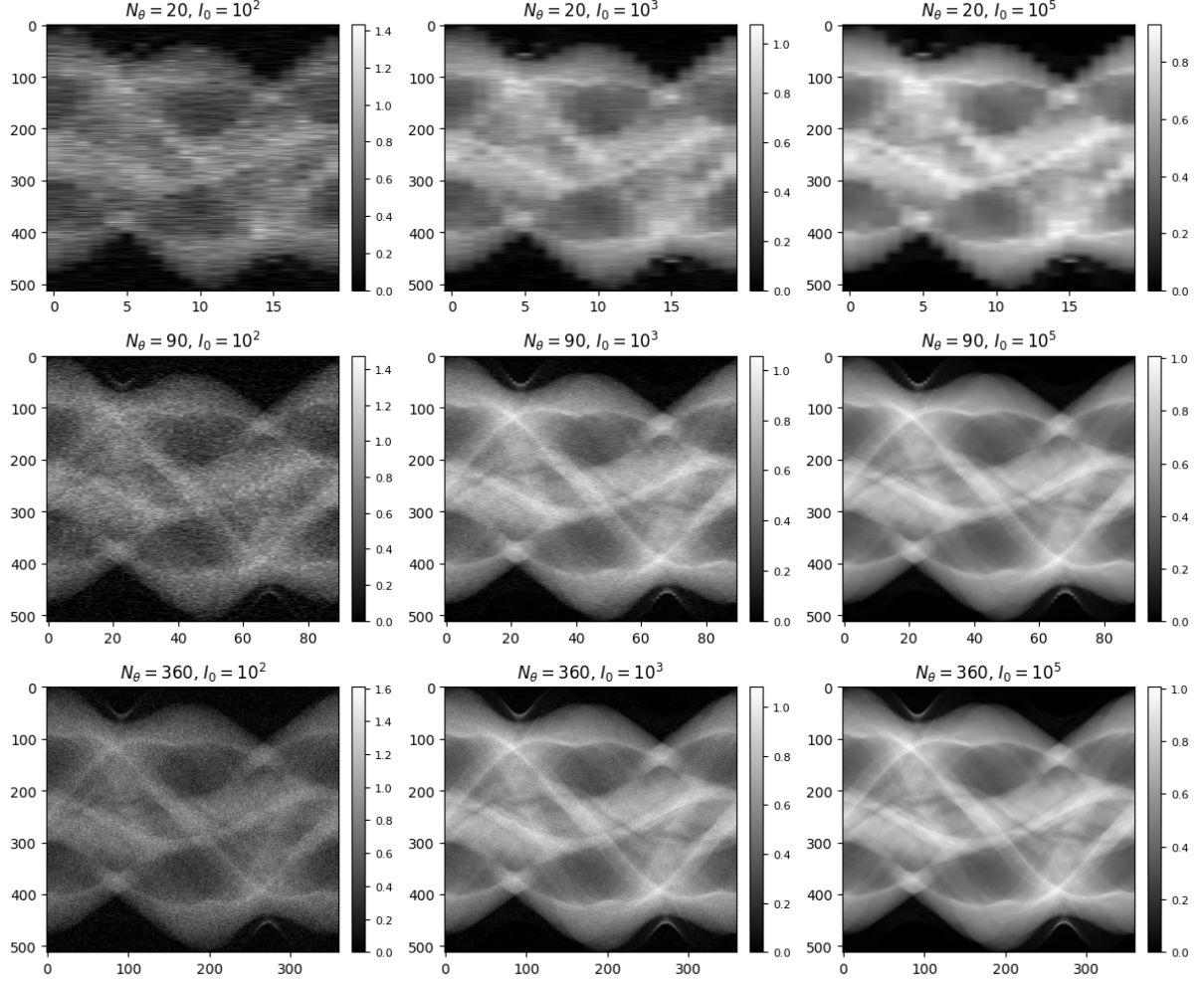


Figure 2: Noisy sinograms across projection count (N_θ) and dose setting (I_0).

Table 1: FBP reconstruction metrics for Exercise 1.1(c); total runtime 2.23 s.

N_θ	I_0	RMSE	PSNR	SSIM
20	10^2	0.0125	-1.97	0.1813
20	10^3	0.0042	7.61	0.1876
20	10^5	0.0011	19.40	0.3408
90	10^2	0.0055	5.16	0.1841
90	10^3	0.0019	14.58	0.2137
90	10^5	0.0004	28.78	0.5741
360	10^2	0.0028	10.99	0.1946
360	10^3	0.0010	19.98	0.2779
360	10^5	0.0001	37.93	0.8789

Table 2: Gradient descent / SIRT reconstruction metrics for Exercise 1.1(c); total runtime 244.14 s.

N_θ	I_0	RMSE	PSNR	SSIM
20	10^2	0.0007	22.69	0.3316
20	10^3	0.0005	26.04	0.5710
20	10^5	0.0005	26.69	0.6834
90	10^2	0.0006	24.10	0.3629
90	10^3	0.0003	29.20	0.6605
90	10^5	0.0003	30.51	0.8173
360	10^2	0.0004	27.03	0.4895
360	10^3	0.0003	30.35	0.7652
360	10^5	0.0003	30.92	0.8552

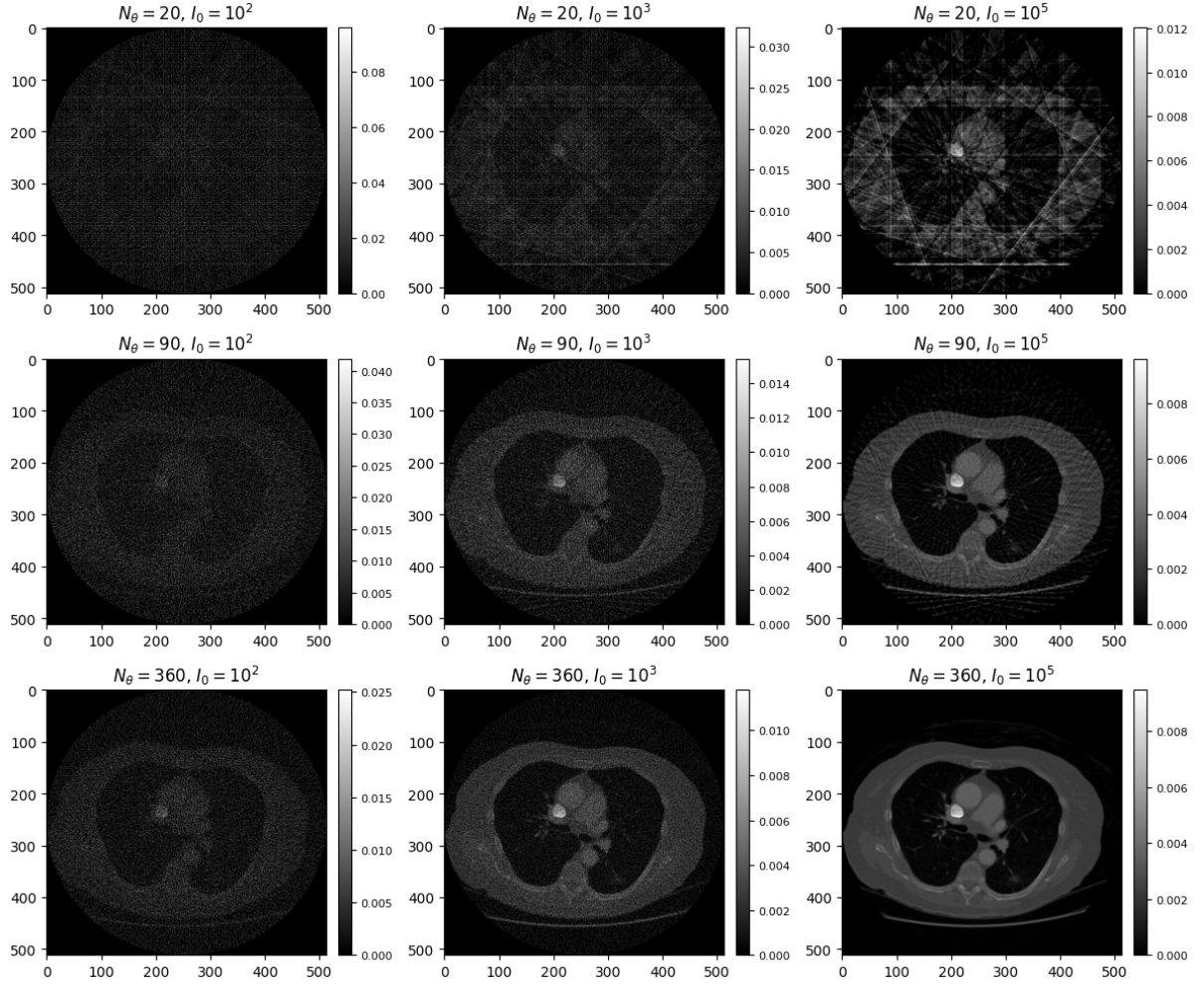


Figure 3: FBP reconstructions across N_θ and I_0 .

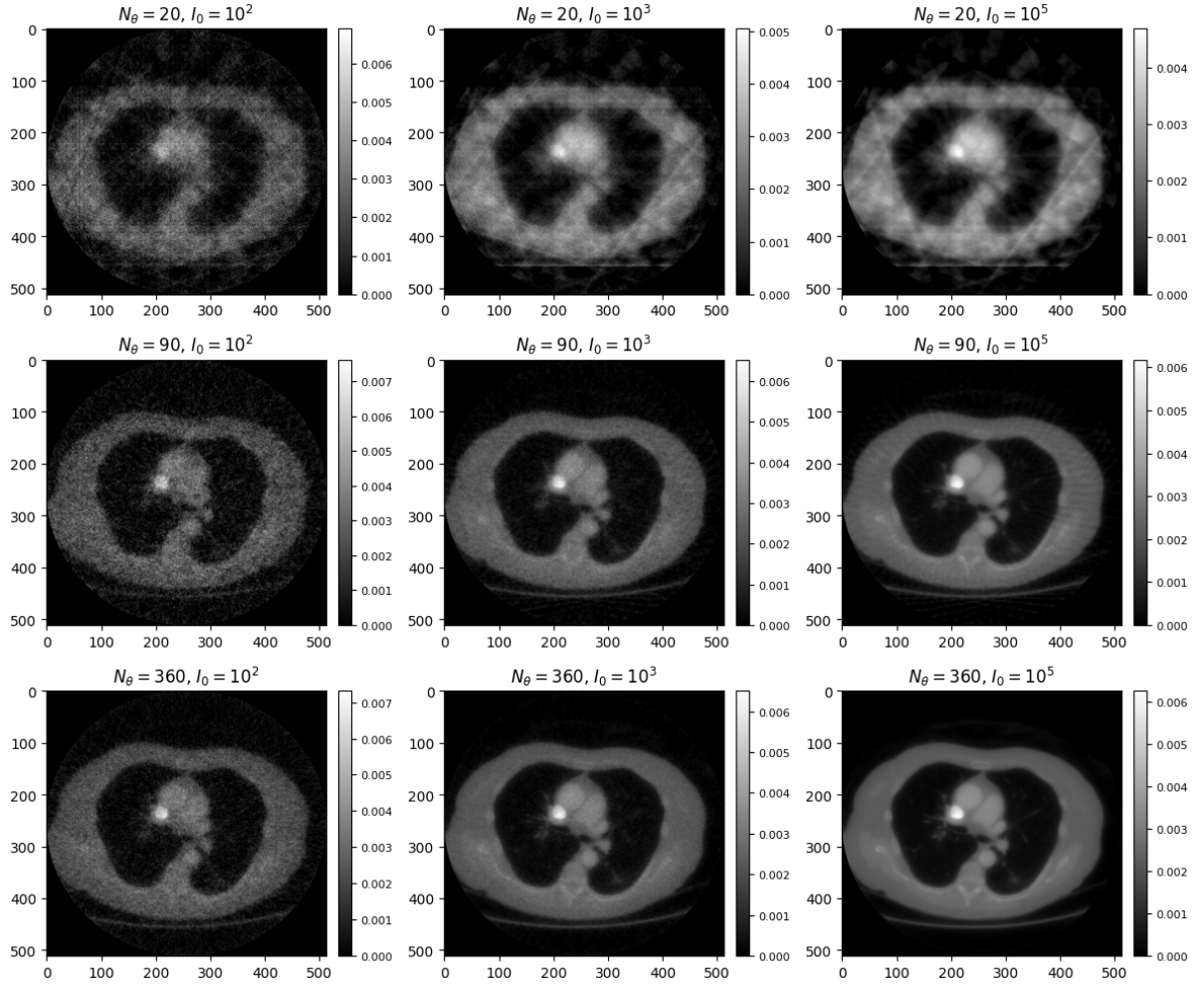


Figure 4: Gradient descent / SIRT reconstructions across N_θ and I_0 .

1.2 Exercise 1.2: Limited angle tomography

(a) Setup and simulations

Figure 5 shows the limited-angle sinograms for fixed $N_\theta = 180$, angular ranges of 40° , 120° , and 180° , and $I_0 = 10^2, 10^3, 10^5$. Higher I_0 reduces graininess, but narrower coverage captures only part of the full sinogram.

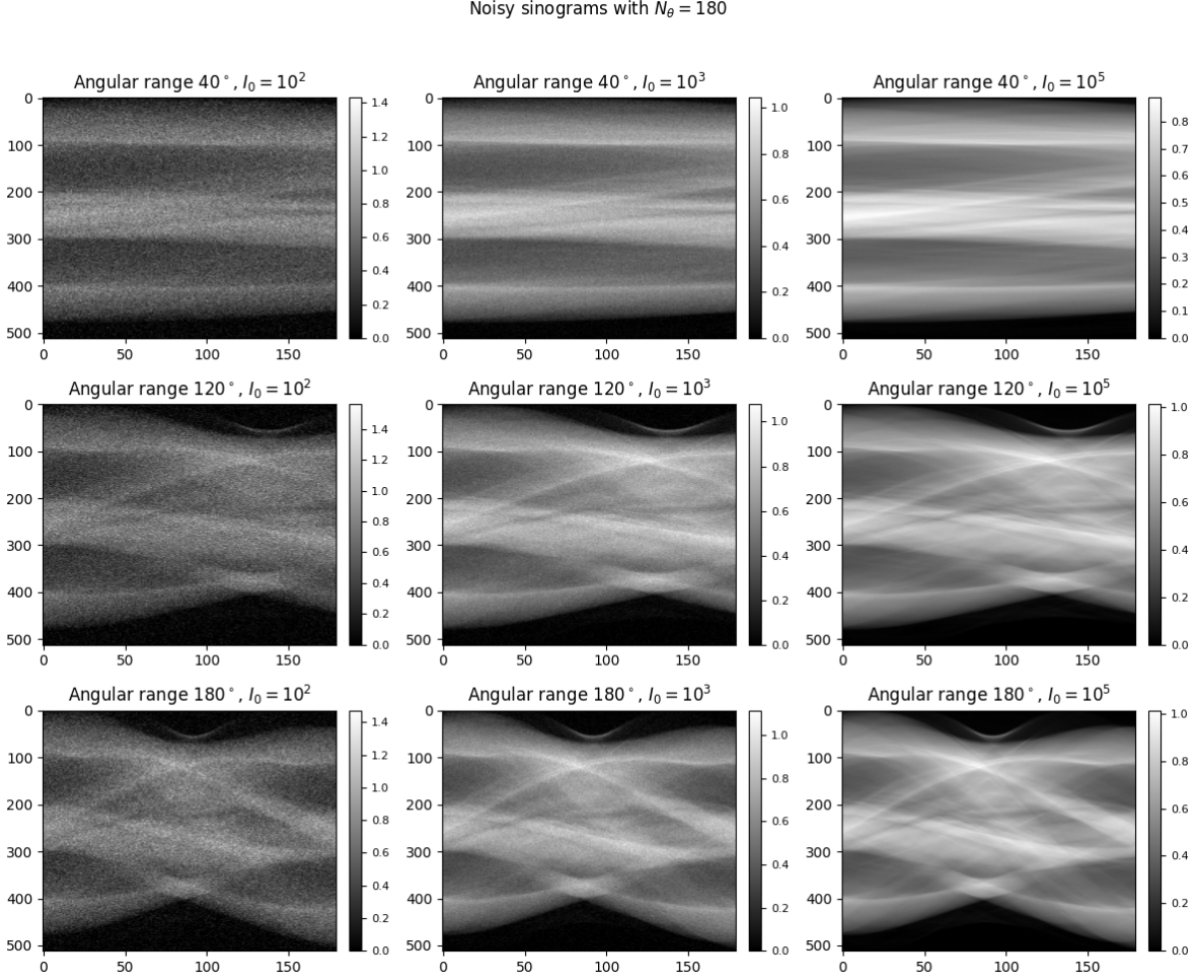


Figure 5: Rows: 40° , 120° , 180° ; columns: $I_0 = 10^2, 10^3, 10^5$, with fixed $N_\theta = 180$.

(b) Observed differences

Figures 6 and 7, with Tables 3 and 4, show that FBP keeps sharper boundaries when coverage is wide and the dose is high, but it degrades much faster as the angular range falls. At 40° long streaks obscure most detail, and even raising I_0 from 10^2 to 10^5 only lifts SSIM from 0.1891 to 0.3979. GD is more stable in the weak cases, with RMSE at 40° , $I_0 = 10^2$ falling from 0.0041 in FBP to 0.0010 in GD, and SSIM at 120° , $I_0 = 10^5$ rising from 0.6826 to 0.7355. In the best data case of 180° , $I_0 = 10^5$, FBP still has the lowest RMSE at 0.0002 versus 0.0003.

(c) Mathematical principle and edge visibility

This behaviour follows the Fourier slice theorem, each projection angle samples one line in the object's Fourier domain, so limited-angle CT leaves a missing wedge of frequencies. As a result, some directions are poorly constrained and streaks appear. An edge can be seen when at least

FBP reconstructions with $N_\theta = 180$

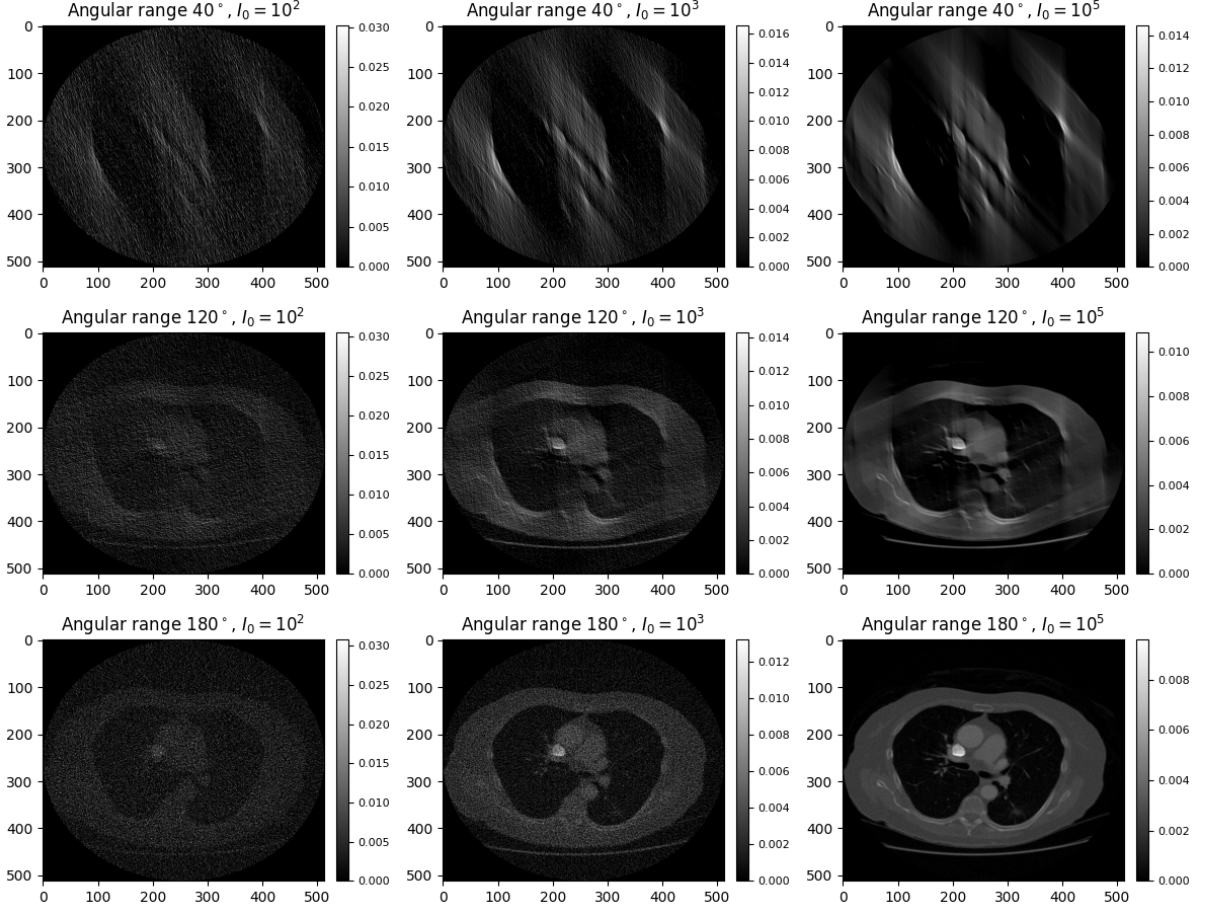


Figure 6: FBP reconstructions for limited-angle tomography with fixed $N_\theta = 180$. Rows correspond to 40° , 120° , and 180° ; columns correspond to $I_0 = 10^2, 10^3, 10^5$.

Table 3: FBP reconstruction metrics for Exercise 1.2(b); total runtime 3.27 s.

Range	I_0	RMSE	PSNR	SSIM
40	10^2	0.0041	7.80	0.1891
40	10^3	0.0019	14.65	0.2461
40	10^5	0.0013	17.42	0.3979
120	10^2	0.0040	8.04	0.1884
120	10^3	0.0015	16.68	0.2444
120	10^5	0.0006	24.15	0.6826
180	10^2	0.0039	8.10	0.1877
180	10^3	0.0014	17.28	0.2383
180	10^5	0.0002	35.51	0.8084

Table 4: Gradient descent / SIRT reconstruction metrics for Exercise 1.2(b); total runtime 292.01 s.

Range	I_0	RMSE	PSNR	SSIM
40	10^2	0.0010	20.20	0.3445
40	10^3	0.0009	20.93	0.5083
40	10^5	0.0009	21.03	0.5622
120	10^2	0.0007	22.95	0.3865
120	10^3	0.0006	24.98	0.6433
120	10^5	0.0005	25.28	0.7355
180	10^2	0.0005	25.26	0.4053
180	10^3	0.0003	29.82	0.7109
180	10^5	0.0003	30.91	0.8543

GD reconstructions with $N_\theta = 180$

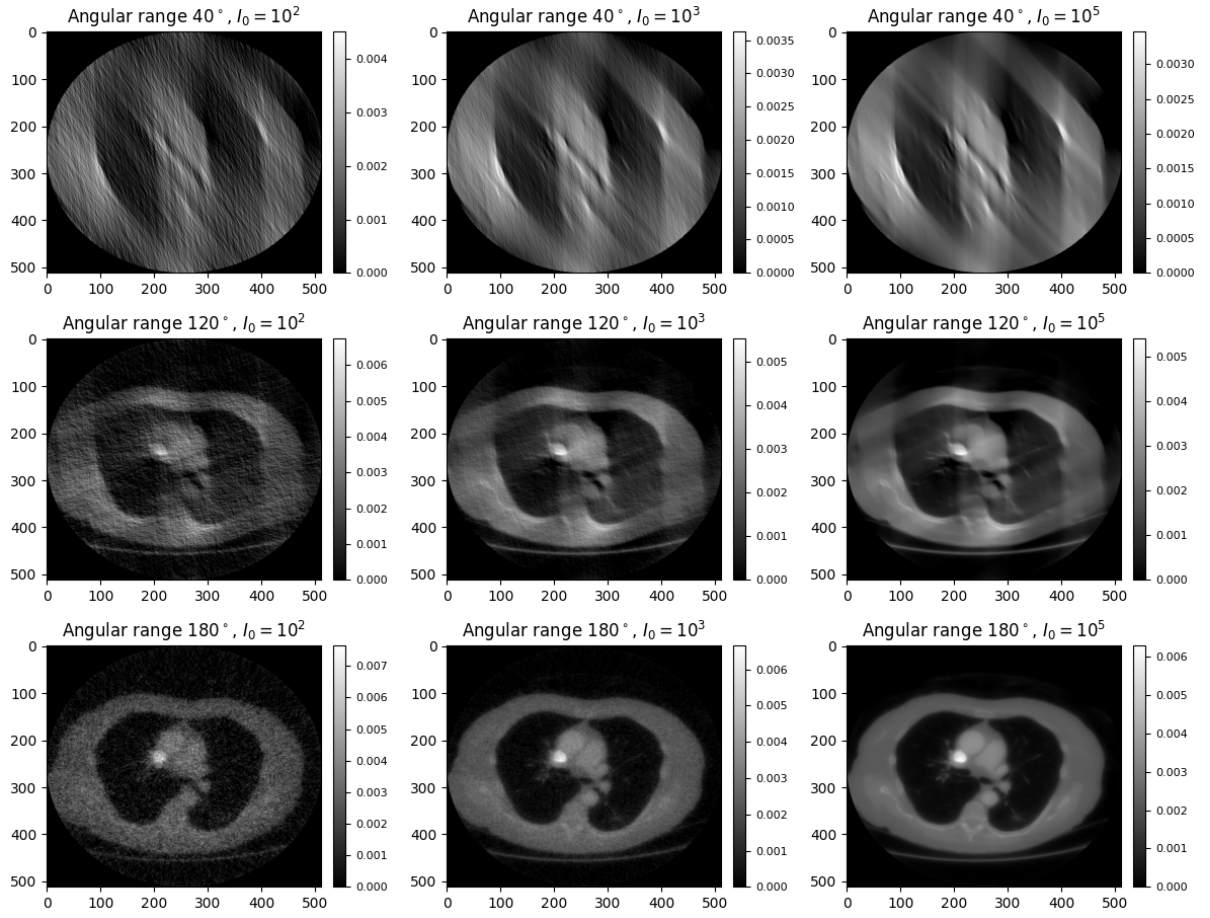


Figure 7: Gradient descent / SIRT reconstructions for the same limited-angle settings as Figure 6.

one measured ray is tangent to it, i.e. when the normal to the edge lies inside the measured angular range (which is at a right angle to measured ray direction). If that normal direction is missing, the edge is blurred in that direction.

1.3 Exercise 1.3: Reconstruction

(a–b) Alternative FBP filters

The standard filter used in FBP is the Ram–Lak filter, a ramp filter that increases the gain on high frequencies to counter the blurring from back-projection. Two common alternatives are Shepp–Logan and Cosine; both apply a window to the ramp so high frequencies are reduced, which lowers noise. Shepp–Logan uses a sinc window and gives moderate suppression, while Cosine suppresses high frequencies more strongly, as seen in Figure 8, so it should reduce high-frequency noise more aggressively.

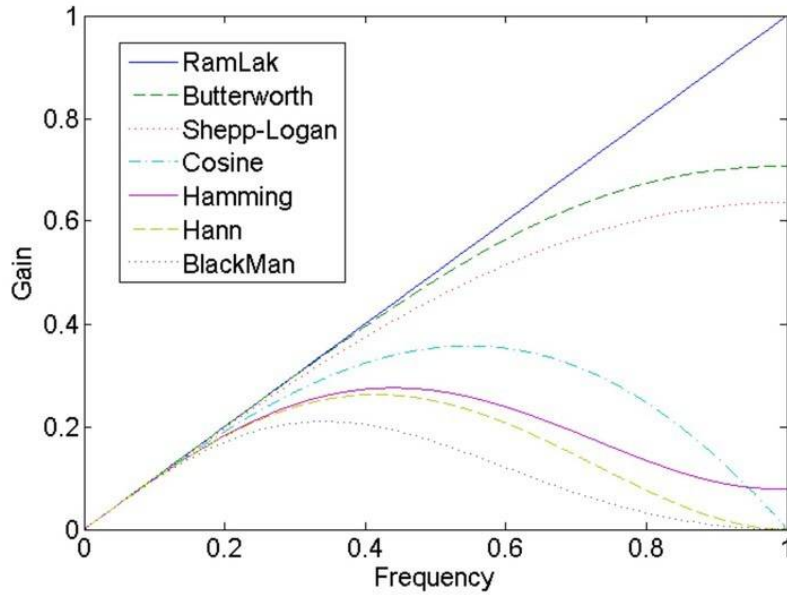


Figure 8: Frequency responses for Ram–Lak and common windowed alternatives, adapted from standard CT filter definitions in [1].

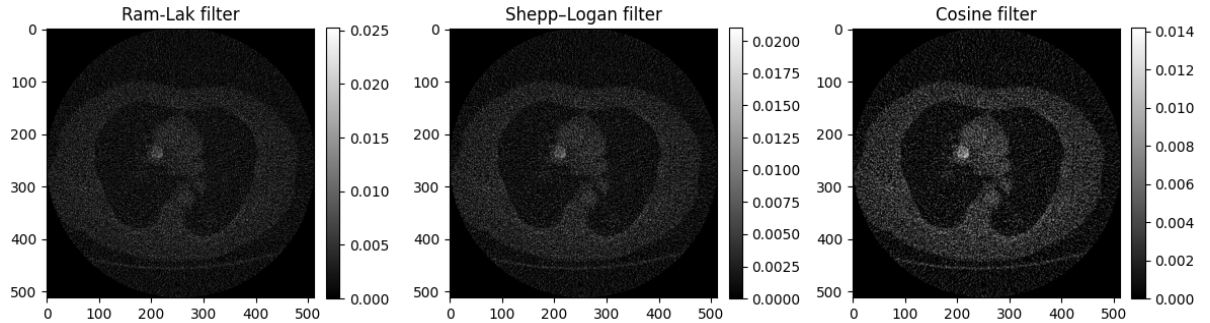


Figure 9: Filter comparison for the low-dose full-angle case $(N_\theta, I_0) = (360, 10^2)$.

In Figure 9, Cosine gives the clearest reconstruction, while Shepp–Logan sits between it and Ram–Lak. Ram–Lak looks less noisy at first, but much of the image is actually just faint apart from a strong central spot, whereas Cosine keeps more detail visible across the object due to a

Table 5: Filter comparison metrics for Figure 9.

Method	RMSE	SSIM
Ram-Lak	0.0028	0.1946
Shepp-Logan	0.0023	0.2012
Cosine	0.0016	0.2251

smaller frequency range. The metrics also follow that pattern, Cosine gives the lowest RMSE at 0.0016 and the highest SSIM at 0.2251, Shepp-Logan is intermediate, and Ram-Lak is worst on both.

(c) OS-SART / ordered subsets vs SIRT

SIRT uses all projections to compute one update per iteration, whereas OS-SART splits the projections into ordered subsets and updates the reconstruction after each subset. This can improve convergence if the relaxation is tuned well, however, in this low-dose case the runtime stayed about the same.

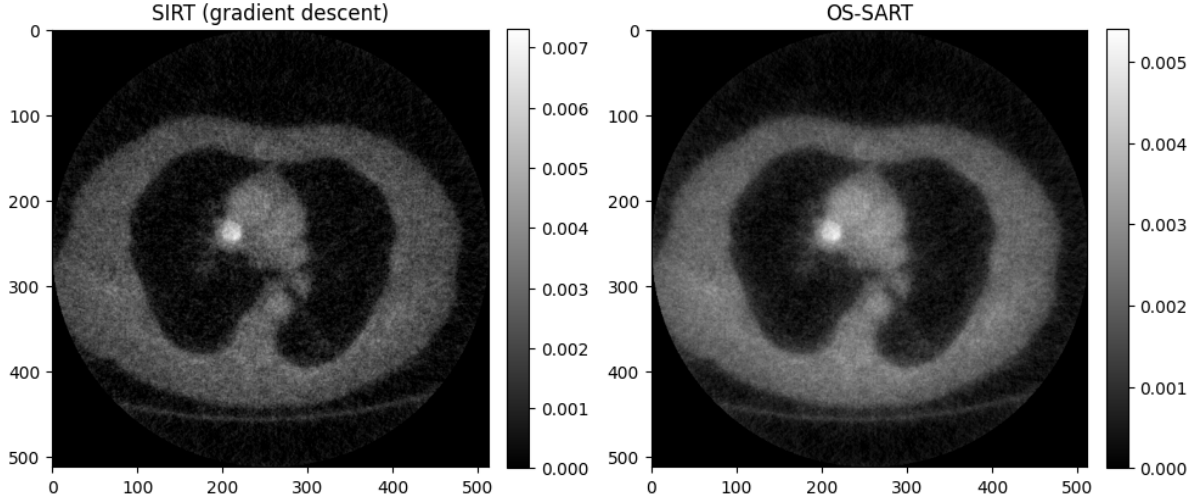


Figure 10: SIRT ($\gamma = 0.001$) and OS-SART ($\gamma = 0.0001$) comparison for $(N_\theta, I_0) = (360, 10^2)$ with 50 iterations and 5 ordered subsets.

Table 6: SIRT and OS-SART metrics for Figure 10.

Method	RMSE	SSIM	Runtime (s)
SIRT	0.0004	0.4895	64.01
OS-SART	0.0005	0.5789	65.20

In Figure 10, SIRT used $\gamma = 0.001$ and OS-SART used $\gamma = 0.0001$. SIRT is a little grainier, while OS-SART has smoother internal regions and a higher SSIM of 0.5789 versus 0.4895 in SIRT, so the overall structure is better preserved. SIRT keeps a slightly lower RMSE of 0.0004 versus 0.0005. Runtime is almost unchanged at about 64–65 s because all 360 projections are still processed each outer iteration, only split into five subsets. So, in this case OS-SART did not give a speed advantage, but it gave the preferred structural reconstruction.

(d) PET vs CT iterative methods

MLEM and OSEM are standard methods in PET because the data is the measurement of photon counts occurring from random decay events, which follow a Poisson model. This means that maximising a Poisson likelihood, as in MLEM and OSEM, is the natural reconstruction method. In CT scans, the transmission data is the measurement of photons that pass through the body after attenuation, this follows the Beer-Lambert law, which after taking the negative log can be represented as a linear system that is effectively solved using SIRT-type methods, although notably low-dose CT can also benefit from more statistical models.

2 Module 2: MRI image denoising

2.1 Exercise 2.1: Visualisation and identifying noise

The k-space data was loaded as an array of shape $(6, H, W)$, so the coil dimension is axis 0. Log-scaled k-space magnitudes are shown in Figure 11. All six images show a bright central spot, with coils 1, 3, 5, and 6 also showing a diagonal streak from the centre, suggesting they all share a similar dominant frequency structure of the knee anatomy. Coil 6 also contains a bright horizontal line in the lower half, which suggests a potential issue with that coil. The edges are noisy across all coils, likely due to thermal noise.

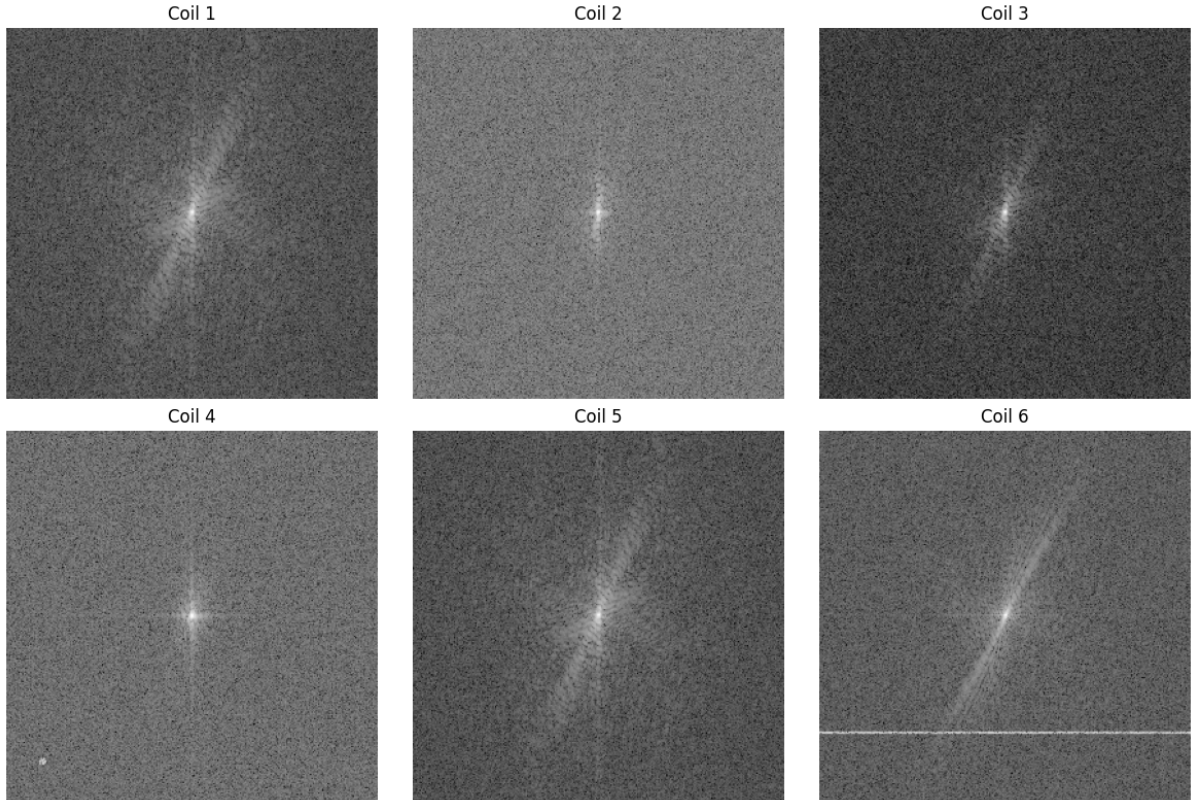


Figure 11: Log-magnitude k-space for each of the six coils.

After the inverse FFT, the magnitude image from Coil 1 in Figure 12 shows the knee anatomy but only over a limited front section, due to the coil's limited distance sensitivity. The phase image is smooth in high-signal regions and random in the background, as expected at low SNR.

The six coil magnitude images in Figure 13 show different sensitivity profiles, consistent with the physical placement of the receiver coils. Coil 6 shows the bottom right of the knee with the

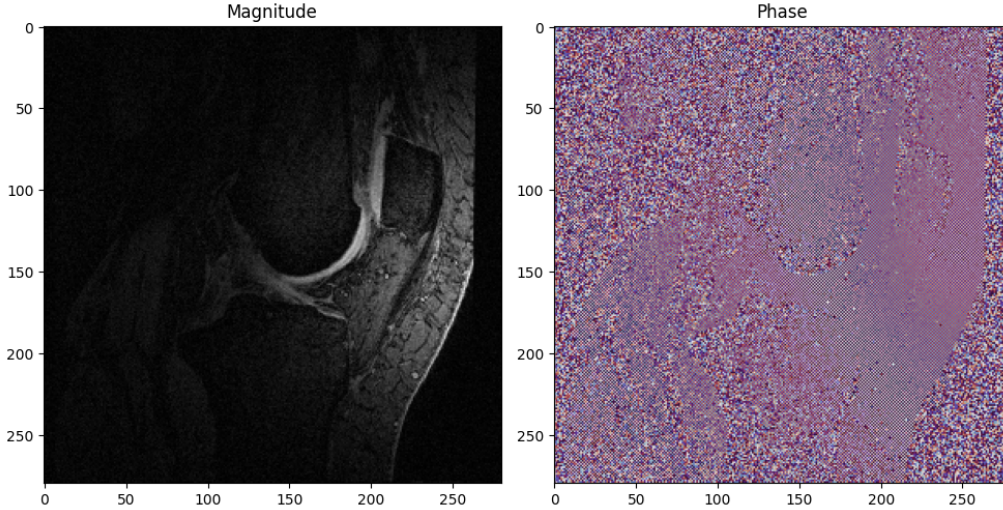


Figure 12: Magnitude (left) and phase (right) from Coil 1 after inverse FFT. Phase is smooth in the anatomy but random in low-signal background regions.

lowest signal of all images, interestingly it also shows weak horizontal bands in the background, which suggests a structured artefact rather than simple random noise.

The root-sum-of-squares (RSS) image, $\sqrt{\sum_c |I_c|^2}$, in Figure 14 gives more uniform coverage of the knee than any single coil and a modest SNR improvement over the best individual images. Normally we would expect a clear SNR gain because in RSS, signal strengths add $\propto N$ coils, whereas the noise adds $\propto \sqrt{N}$, this lack of increase is likely because of the inclusion of the lower SNR coils such as coil 2, 3, and 4.

2.2 Exercise 2.2: Removing noise

(a) Three image-space denoisers

Three coil-wise denoisers were tested on the reconstructed images from (Figure 13), these methods were Gaussian filtering with $\sigma = 1.0$, bilateral filtering with $\sigma_{\text{spatial}} = 3.0$, and wavelet denoising using BayesShrink. The differences are most obvious in coils 2 and 4, which are the noisiest, whereas coils 1 and 5 already have a relatively strong signal and therefore clear images even before the denoising. Coils 1 and 5 produce identical metrics throughout, suggesting they share the same underlying k-space data.

Table 7: Per-coil denoising metrics for the three image-space methods. SSIM is measured against the noisy input, so it reflects structural preservation rather than absolute image quality.

Coil	SNR _{orig}	Gaussian		Bilateral		Wavelet	
		SNR	SSIM	SNR	SSIM	SNR	SSIM
1	5.60	17.89	0.775	37.20	0.735	10.37	0.933
2	2.15	7.66	0.401	6.52	0.672	13.52	0.267
3	3.00	10.75	0.540	16.48	0.626	9.85	0.604
4	2.07	6.81	0.409	6.94	0.633	13.57	0.260
5	5.60	17.89	0.775	37.20	0.735	10.37	0.933
6	2.05	5.40	0.818	11.76	0.809	3.02	0.935

Figures 15, 16, and 17, with Table 7, show that Gaussian is the most uniform smoother. It removes a large amount of the fine noise from coils 2 and 4 (raising coil 2 from SNR 2.15 to 7.66), but it also softens and brightens boundaries of bone and soft tissue in coils 1, 3, and 5, making the details inside the knee more blurry. SSIM stays below 0.82 across all coils, so despite

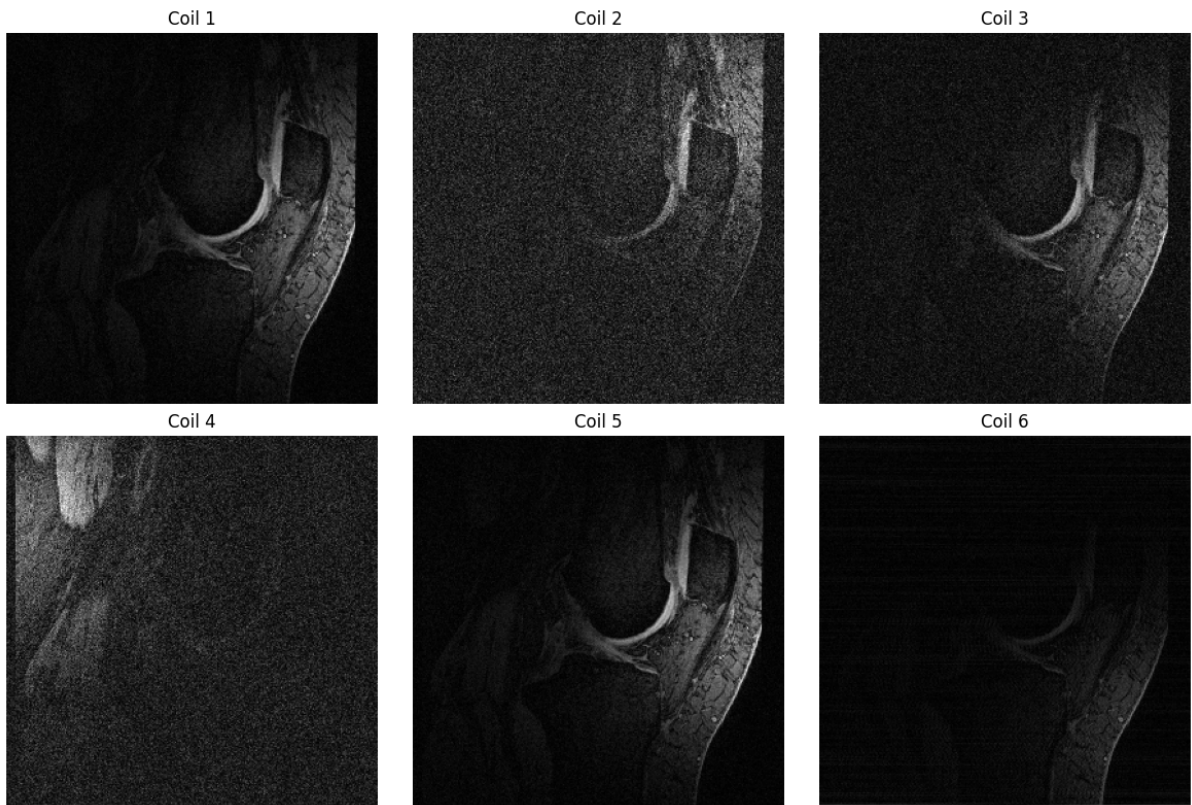


Figure 13: Magnitude images in image space across all six coils.

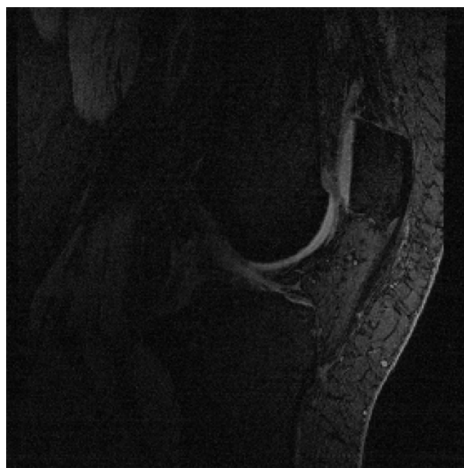


Figure 14: RSS combined magnitude image. Full anatomical coverage with improved SNR compared to individual coils.

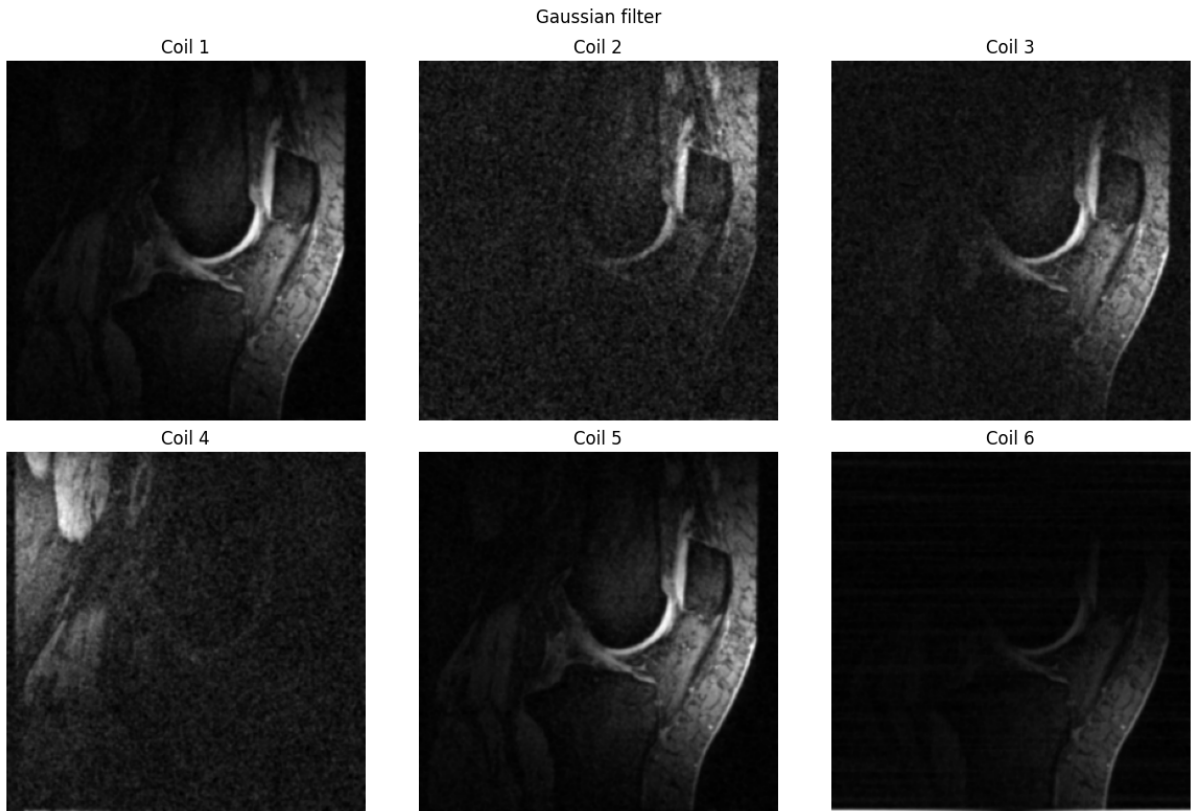


Figure 15: Gaussian denoising applied per coil.

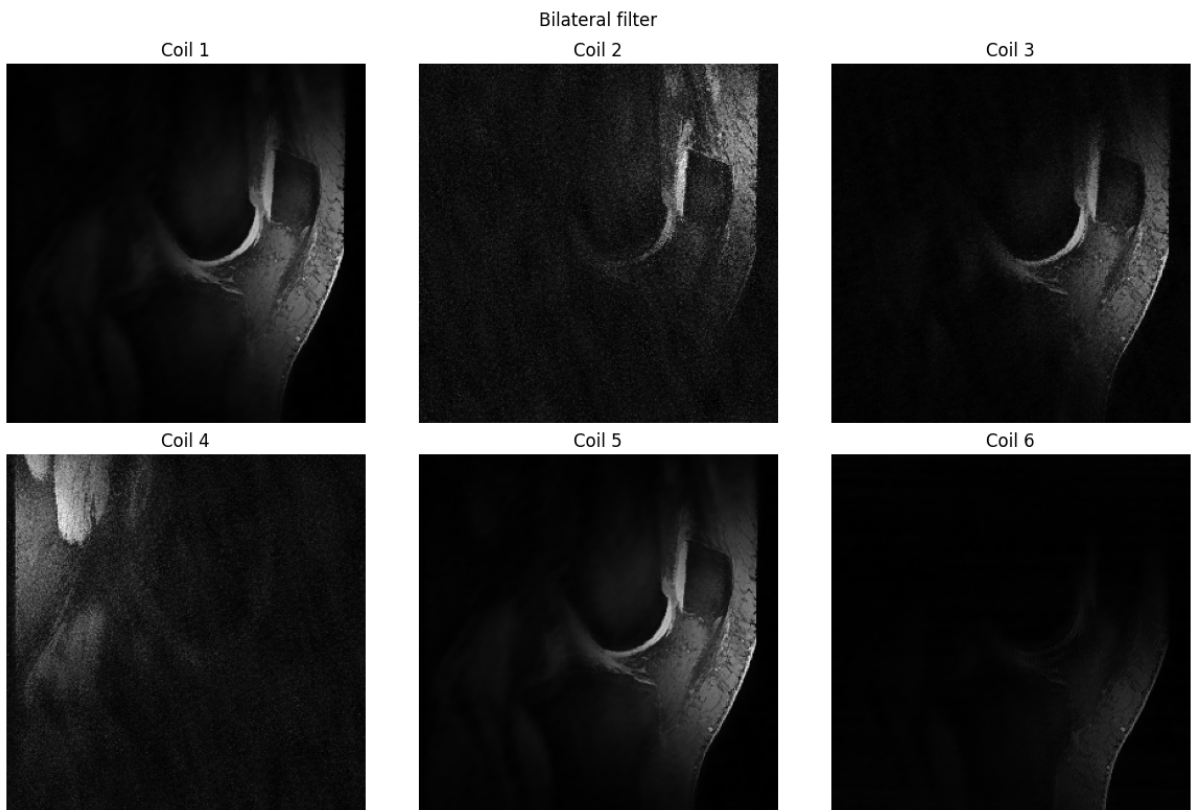


Figure 16: Bilateral denoising applied per coil.

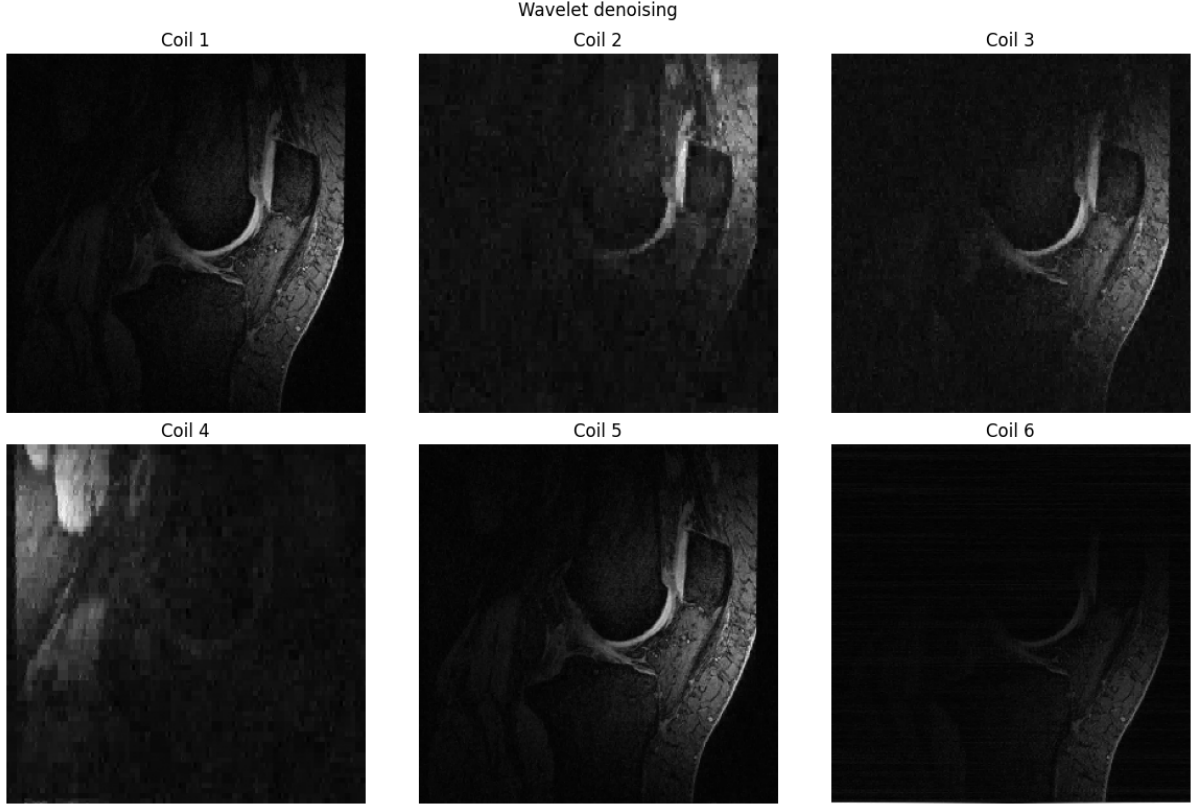


Figure 17: Wavelet denoising applied per coil.

some noise reduction on coils 2 and 4 the filter provides limited overall value.

Bilateral filtering is much more effective than the Gaussian denoising, it preserves the edges of the tissue better whilst still giving a useful noise reduction. Even for the more noisy coils 2 and 4, interior details and edges are very clearly preserved, although the lesser varying sections become more flattened creating jagged shapes. On the cleaner images the bilateral filter manages to almost entirely remove background noise however, if the details inside the knee are not high contrast it tends to blur them together. It also manages to remove the light banding across the image in coil 6. Generally bilateral seems to provide a very good compromise between smoothing and anatomical sharpness, reaching SNR 37.20 on coils 1 and 5 compared to 17.89 from Gaussian (Table 7).

Wavelet denoising is more adaptive to the image itself, which helps in some coils but makes the behaviour less consistent overall. In coils 1 and 5 the main structures are almost entirely unchanged, but in coils 2 and 4 the output becomes much blockier and the contrast looks less natural, despite wavelet actually achieving the highest denoised SNR on these coils (13.52 and 13.57, roughly double bilateral's values). The metrics and visual feel disagree here, the noise is suppressed but the blockiness obscures finer detail, with SSIM ranging from 0.260 on coil 4 to 0.935 on coil 6. Overall, bilateral filtering is the strongest of the three options here because it removes a useful amount of noise without losing as much anatomical detail as Gaussian or producing the patchier appearance seen with wavelets.

(b) k-space Butterworth low-pass

A Butterworth low-pass filter ($D_0 = 30$, $n = 2$) was applied to the first coil's k-space data before transforming it into image space. Unlike the image-based methods which work based on pixel intensities, this approach suppresses noise by cutting off high frequencies in the Fourier domain.

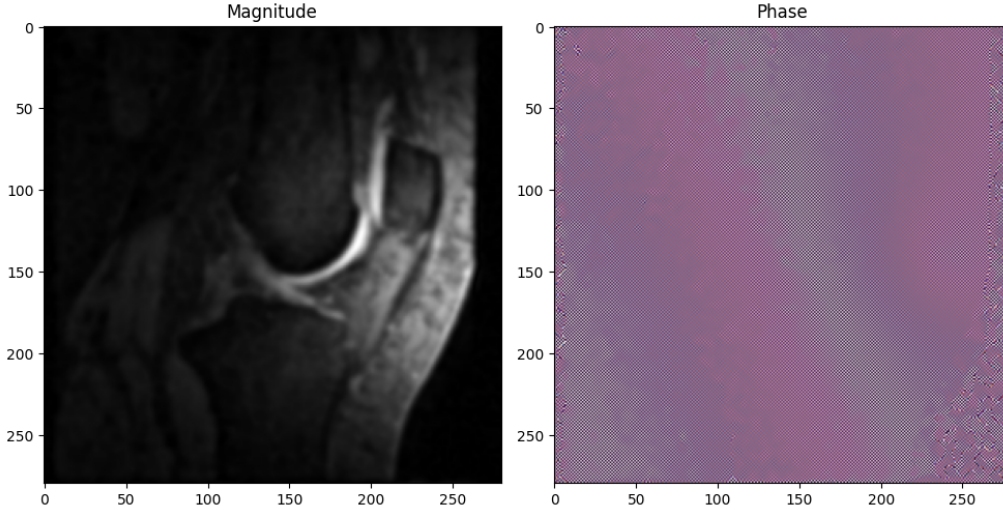


Figure 18: First-coil magnitude and phase after applying a Butterworth low-pass filter in k-space.

As seen in Figure 18, this denoising via frequency cut-off produces a very different result compared to the previous methods. The magnitude image is strongly blurred, even more so than Gaussian, and the previously sharp anatomical boundaries of the knee are lost. This likely occurs because the Butterworth filter struggles to distinguish between high-frequency random thermal noise and the high-frequency data that occurs at the edges of the tissues, meaning it suppresses both equally. In contrast, "edge-aware" spatial filters like the bilateral and wavelet methods much more successfully preserved these boundaries whilst also removing the local noise.

The change in the phase image from Figure 12 reflects low-pass transformation as well, the previously high variance random background phase has been replaced by a much smoother image with the same wave pattern occurring as in the smoother parts of the first image. This only further demonstrates the complete loss of independent high-frequency components.

(c) Denoising on combined image

Bilateral filtering was chosen for the combined image based on its per-coil performance (Table 7), and the denoised coils were combined with RSS. Figure 19 shows a fairly modest but useful change from the original combined image in Figure 14. The dark background is less noisy, and the larger soft-tissue regions inside the knee look more uniform. At the same time, the bright curved interior lines and the outer soft tissue edge on the right remain clearly defined, so the main anatomy is still easy to follow. Some finer internal texture is slightly weaker than before, but the image does not look heavily smoothed.

(d) Comparison of denoising methods

Across the methods tested here, bilateral filtering gives the best compromise both visually and quantitatively, with the highest denoised SNR on four of six coils. Wavelet technically does better than it on the two noisiest coils (2 and 4), but the blockiness outweighs the better metrics as discussed above. Gaussian smoothing reduces random grain, but it softens the internal edges and nearby transitions from tissue to bone too much. Wavelet denoising keeps more edge information than Gaussian, but in the noisiest coils it introduces a blockier high-contrast texture that looks less anatomical, and SSIM drops as low as 0.260. Butterworth filtering is the weakest result visually, because removing high frequencies in k-space also removes genuine edge information. The magnitude image becomes noticeably blurred and the phase image is overly smooth. The bilateral-plus-RSS image therefore seems to be the strongest final result,

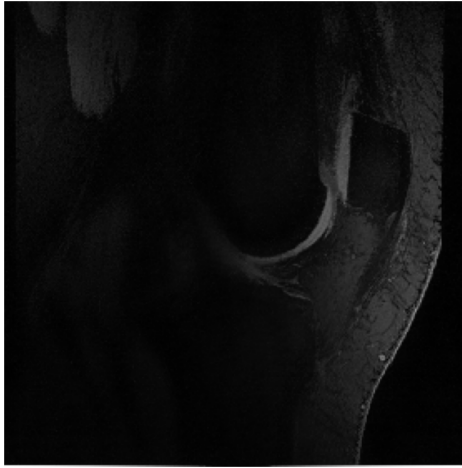


Figure 19: RSS image formed after bilateral denoising of each coil magnitude image.

even managing to remove the structured banding seen in coil 6. A good next step would be a non-local means method or applying denoising post-combining on data, as these are more likely to suppress repeated noise patterns without flattening the main anatomy.

3 Module 3: Image segmentation methods

3.1 Exercise 3.1: Critical comparison of methods

Selected Papers

Processing-based paper: Zhou, Z., Wan, H., Zhang, H., Chen, X., Wang, X., Song, L. and Zhang, T. (2022). *Segmentation of Spontaneous Intracerebral Hemorrhage on CT With a Region Growing Method Based on Watershed Preprocessing*. *Frontiers in Neurology*, 13, 865023 [2].

ML/DL paper: Yu, H., Hu, C., Wang, J., Ma, N., Li, H. and Yu, N. (2022). *A Robust Deep Learning Segmentation Method for Hematoma Volumetric Detection in Intracerebral Hemorrhage*. *Stroke*, 53(1), 167–176 [3].

(b) Paper summaries

The Zhou et al. paper proposes a faster method for identifying intracerebral hemorrhages, specifically measuring bleed volume using CT. They use a semi-automatic pipeline where a marker-based watershed step splits the brain tissue into regions based on "filling" around local intensity minima. A doctor then selects which regions fall inside the hemorrhage, setting them as seed points. A two-pass region growing algorithm is then applied, the first pass uses a fixed intensity threshold of 0.30 which gives a rough outline of hemorrhage pixels, the second pass refines it with Otsu's method, which picks an adaptive cutoff by maximising variance across regions. When a hemorrhage is close to the skull, a separation step is used to remove the bone via thresholding (brightness cutoff) and morphological operations to prevent leakage. On 55 patients they report Dice 0.93 ± 0.02 (overlap with doctors hand-drawing) and sensitivity 0.92 ± 0.04 (% bleed caught).

Yu et al. approach the same problem with DR-UNet, a U-Net where dimensionality reduction modules in both the encoder and decoder compress the parameter space, which lowers the computational cost whilst not seriously harming its segmentation ability. The network was trained on around 5,000 hematoma slices with doctor-drawn outlines from 512 patients and then benchmarked against a standard U-Net, fuzzy C-means (pixel clustering method), and active

contours (boundary defining algorithm). DR-UNet achieved Dice 0.861 ± 0.139 on the internal test set and 0.874 ± 0.130 on a prospective set of 50 patients collected later at the same hospital, and volume correlation was $R^2 = 0.9979$ against the doctors hand-drawn measurements. The similar Dice on the prospective set suggests the network generalises to new patients rather than overfitting to the training cases. They also validated on trickier sub-groups with no significant drop.

(b) Critical comparison

The most immediate difference is automation. Zhou’s pipeline requires a human to review the watershed output and select seed regions that lie within the hemorrhage region before starting the region growing step, so it can’t process scans without a human. The DR-UNet however is fully automatic, a CT slice is fed in and the software produces a mask. This speed is important in a clinical setting where stroke patients need very quick hemorrhage volume assessments, although the semi-automatic step does also give the doctor a chance to catch errors before the final segmentation is output, which is a form of built-in quality control that the fully automatic approach lacks.

Focussing on accuracy metrics, Zhou reports Dice 0.93 versus 0.861–0.874 for DR-UNet, but these numbers aren’t directly comparable. Zhou tested on 55 patients from one hospital, all with a specific type of hemorrhage making the test set fairly homogeneous. Yu’s evaluation covered 512 patients internally and 50 in a later prospective cohort from the same hospital, which included a wider range of hemorrhage types and sizes. The fact that Yu’s prospective Dice (0.874) is actually slightly higher than the internal one (0.861) suggests the model generalises to new patients over time, though both sets are from one hospital so external performance remains untested. Zhou’s higher Dice potentially reflects the smaller case variation as much as the efficacy of the method itself.

The upfront requirements for each method are also quite different. Zhou’s pipeline requires no training data, the thresholds and morphological parameters are hard-coded, so any hospital could run it on their own scans immediately. DR-UNet needed 512 patients of manual annotation to train, which is a substantial effort. For very rare case types or settings where large annotated datasets don’t exist, Zhou’s non-ML approach has a clear advantage.

From the tests DR-UNet does better on atypical cases. Zhou et al. only evaluated standard ICH types, the one edge case addressed was the skull-boundary leakage mentioned earlier. Yu et al. explicitly tested on smaller, harder case groups such as irregular hematomas, subdural and epidural hemorrhages, as well as three larger volume groups, with accuracy holding up across all of them. Since in reality clinical hemorrhages often vary considerably in shape and location, this broader validation is important.

The computational constraints of the papers are slightly counterintuitive. Zhou’s non-ML pipeline runs on a standard CPU in MATLAB at 18 seconds per patient, needing no special hardware. DR-UNet needs a GPU, but inference is much faster only taking about 1.2 seconds per patient, and the computationally expensive training phase is a one-off cost.

Overall the classical pipeline seems better suited to settings with limited data and relatively uniform case-types, while DR-UNet is the stronger choice when annotated training data is available and patients are presenting with more varied hemorrhage types. Looking forward Yu’s method is likely stronger as more training data becomes available.

(c) Proposed improvements

As discussed above Zhou et al.’s largest constraint in a clinical setting is the manual seed region selection step. A human operator has to review the watershed output and pick which regions lie inside the hemorrhage before the region growing step can be started. This prevents

batch processing, introduces human error, and makes the method unsuitable time constrained environments like a stroke triage.

One fix for this would be to replace the manual selection with a small CNN trained to classify watershed candidate regions as either hemorrhage or background. Because the network only makes a binary decision on small already-sectioned regions rather than segmenting whole images, it means that a shallow architecture (3–4 convolutional layers, global average pooling, sigmoid output) would likely be effective. The training data requirement is modest, Zhou’s 55-patient dataset already contains hundreds of watershed regions that could be labelled as positive or negative seeds, and augmentation (rotation, flipping, intensity jitter) would expand this. Precise boundary setting is still handled by the two-pass region growing step with Otsu refinement, so the segmentation choices remains clear rather than relying on a black-box prediction, moreover a clinician could also override the CNN’s seed choices if needed.

A second improvement surrounds the evaluation metrics. Zhou tested on 55 patients from one hospital, one scanner type, and only on standard hemorrhage types. Including validation from several hospitals with different CT scanner manufacturers and different cases would make clear whether the hard-coded thresholds (0.30 first-pass cutoff) generalise well or require calibrating on each scanner.

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